



Daffodil
International
University

ASSIGNMENT

Course Code: CIS216L

Course Title: Object Oriented Programming

Topic Name: Java

Submitted To

Name: MD. Nasimul Kader

Designation: Assistant professor

Department of Computing and Information System

Daffodil International University

Submitted By

Name: Sharyara Tabassum Sadia

ID: 253-16-050

Section: B

Semester: 4th

Department of Computing and Information System

Daffodil International University

Date of Submission: 28/09/2026

Introduction to Java Programming

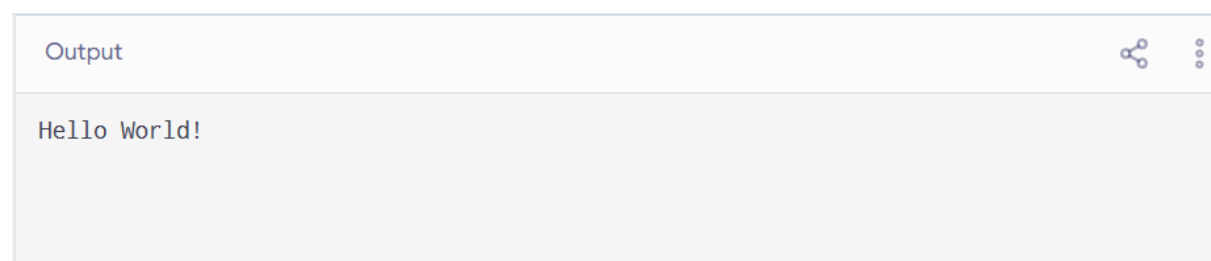
Java is a widely used, high-level, class-based, object-oriented programming language designed by Sun Microsystems in 1995. It operates on the "Write Once, Run Anywhere" (WORA) principle, meaning compiled Java code can run on all platforms that support JVM without needing recompilation.

Core Features of Java

- 1.Object-Oriented Architecture:** Java organizes software around objects and classes rather than actions and logic, facilitating code reusability and modularity.
- 2. Platform Independence:** Java source code is compiled into platform-neutral bytecode, which executes on any system equipped with a Java Virtual Machine (JVM).
- 3.Robustness & Security:** Strong memory management, automatic garbage collection, and strict exception handling make Java applications highly stable and secure.
- 4.Multithreading Support:** Java has native capabilities to execute multiple threads concurrently, maximizing CPU performance.
- 5. Comprehensive Standard Library (API):** Java comes with an extensive rich library offering built-in functions for networking, database connectivity, data structures, and input/output handling.
- 6.Enterprise Framework Support:** Frameworks such as Spring, Hibernate, and Jakarta EE simplify building secure and enterprise-grade software.
- 7.Scalability & Maintainability:** Structured class hierarchies allow large-scale software projects to scale smoothly and remain maintainable over time.

Code-01

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello World!");  
    }  
}
```



Explanation of How It Works:

- 1. public class HelloWorld:**

- Java-te shob code kono nah kono class-er bhetore thakte hoy.
- Ekhane HelloWorld name ekta public class toiri kora hoyeche. File-er naam-o HelloWorld.java hote hobe.

2. **public static void main(String[] args):**

- Ei line-ti hocche Java program-er main entry point, jekhan theke execution shuru hoy.
- **public:** Ei method-ti jeno jekono jayga theke accessible hoy.
- **static:** Class-er kono object toiri na korei jeno method-ti run kora jay.
- **void:** Ei method theke kono value return hobe na.
- **String[] args:** Command-line arguments hold korar jonno ekta String array.

3. **System.out.println("Hello World!");:**

- **System:** Java-er built-in class.
- **out:** Output console stream.
- **println():** Console-e text print kore new line-e jawar method.
- Quotation-er bhetor thaka "Hello World!" text-ti screen-e print hobe.

Code-02

```
public class Daffodil{
    public static void main(String[] args) {
        {
            System.out.println("This is inside the block.");
        }

        System.out.println("This is outside the block.");
    }
}
```

Output



```
This is inside the block.
This is outside the block.
```

Explanation of How It Works:

1. `public class Daffodil`

- **public:** An access modifier that makes this class accessible from any other class.
- **class:** The keyword used to declare a class in Java.
- **Daffodil:** The name of the class. The outer curly braces { ... } enclose all the contents belonging to this class.

2. `public static void main(String[] args)`

- This is the Main Entry Point where the execution of the Java application begins.
- **public:** Allows the JVM to execute this method from outside the class.
- **static:** Enables the JVM to call this method without creating an instance/object of the class.
- **void:** Indicates that this method does not return any data.
- **main:** The standard name recognized by the JVM as the starting method.
- **String[] args:** Stores command-line arguments passed to the program as an array of strings.

3. Standalone Inner Block { ... }

- In Java, curly braces {} can group statements into an independent block even without conditional statements.
- **System.out.println("This is inside the block.");**: Executes unconditionally inside this inner block and prints the text to the console.
- **Scope Rule:** Variables declared inside this block are local to it and cannot be used outside the closing brace }.

4. `System.out.println("This is outside the block.");`

- Located outside the standalone inner block but still inside the main method.
- It executes sequentially after the inner block finishes and prints its text to the console.